

BEHAVIORAL COLLAPSE IN AI-GENERATED HARNESS POPULATIONS

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ABSTRACT

Populations of automatically generated LLM harnesses underpin routing and value prediction, which presuppose that population members *disagree* on outcomes. We audit this premise on BIRD text-to-SQL. A Phase-I discovery study found a standard harness-evolution population syntactically diverse yet behaviorally collapsed — 36 of 91 harness pairs identical on every task, code similarity unassociated with outcome disagreement — and traced the failure by execution-trace audit to mechanisms described but never executed. A hand-written positive control showed the routing headroom exists (7.3 pp); the failure was on the generation side. A pre-specified confirmatory factorial study (strategy forcing $\times$ conformance gate; six builders, three seeds; 1,169 questions from never-opened databases; official BIRD scorer) then revised the story twice. First, collapse proved *protocol-sensitive*: under a clean equal-budget protocol, free ungated generation already produced headroom-rich populations (8.6 pp oracle headroom, 31% union repair). Second, the intervention showed *no clear improvement* (-0.16 pp, CI $[-0.90, +0.82]$); the apparent negative gate effect was substantially size-confounded (-0.30 pp at matched K , CI spanning zero) and the bare-outcome noise floor (14.5% flip rate at temperature 0) exceeds the effects in question. The stable conclusion is a measurement mandate: report outcome disagreement, repair, harm, and headroom before routing over generated populations — and audit the generation protocol, since the collapse we found was invisible to every code-level check the pipeline had passed.

1 INTRODUCTION

The fastest-growing lever on a frozen LLM is no longer weights but the *harness* — the program of prompts, tools, and control flow that wraps the model for a task. A growing line of “AI4AI” systems generates this scaffolding automatically: LLM builders propose, edit, and evolve harness code. The implicit premise is that when the generated code differs, the *behavior* differs — that a population of thirty harnesses is thirty ways of answering, not one way written thirty times. Routing and value prediction over such populations presuppose exactly this.

Phase I (discovery): the premise fails, and for identifiable reasons. On BIRD text-to-SQL (Li et al., 2023) with a frozen mid-tier target, we measure what the candidates of a standard harness-evolution loop actually *do*, not what their source files say (Figure 1). Every candidate passes a code-difference check, yet among the 12 generated candidates (66 candidate pairs) average pairwise outcome disagreement is **3.8%** and **21 pairs are behaviorally identical**; counting the shipped *react* harness and *bare* baseline as well (91 pairs), 36 pairs return identical correctness on every task, and pairwise code similarity shows no observed rank association with outcome disagreement (Spearman $\rho = -0.01$). The shipped *react* variant reproduces *bare*’s final SQL byte-for-byte on every audited task. Execution-trace auditing resolves “different code” into three failure classes: mechanisms *described* but absent from the control flow, feedback with no causal path to the output, and genuinely multi-turn paths that are broken (§3). We call this **behavioral collapse**.

Collapse is not a property of the task: four hand-written harnesses with forcibly distinct execution paths deliver 7.3 pp oracle headroom on the same target and judge — the headroom routers need is real; the generation side fails to populate it. We also document a second, more primitive failure mode in one deployed pipeline: a widely used optimize loop emitted candidates *byte-identical to*

the baseline source. This is not behavioral collapse — no distinct population was generated at all — and we report it as a *generation-null* (M0) failure, kept strictly separate from the collapse claim, which we support only on genuinely distinct-source populations.

Phase II (confirmatory): a pre-specified factorial. Phase I suggested that behavioral collapse can be repaired by generation-side interventions, but its design — two nested contrasts, three builders, unlogged early runs — could not separate protocol from builder effects, its evaluation set overlapped development items, and its judge was a conservative proxy. We therefore froze a confirmatory protocol before any confirmatory outcome was observed: a 2×2 factorial over *strategy forcing* (assigned mechanism vs. free invention) and a *mechanism-conformance gate* (counterfactual trace verification vs. neutral validity only), crossed with six builder models and three seeds, evaluated on 1,169 questions from nine databases never opened during development, scored by the official BIRD judge, with the primary contrast (forced+gated vs. free+ungated populations) analyzed as paired continuous effects with hierarchical bootstrap intervals (§4). The conformance instruments are calibrated against hand-written positive controls and known-negative impostor harnesses before any candidate is scored.

The confirmatory answer revises the story twice: the Phase-I collapse did not reproduce under the clean protocol (free generation was already headroom-rich), and the intervention itself was null within measurement precision — the apparent negative gate effect was substantially size-confounded and not Holm-significant, and the API nondeterminism noise floor exceeds the sub-pp effects in question (Section 5).

Our contributions:

1. **An outcome-matrix measurement of behavioral collapse** in AI-generated harness populations, with an execution-trace taxonomy explaining why code diversity fails to become behavioral diversity, and a clean separation between genuine collapse and generation-null failures.
2. **Calibrated generation-side instruments and a pre-specified factorial study:** a counterfactual conformance suite and self-specified behavioral contracts, both calibrated on positive and negative controls; a 2×2 design with equal raw generation budgets that isolates the combined intervention and its components on uncontaminated data under the official scorer.
3. **A boundary result for routing:** on the Phase-I populations, open-set harness-value prediction does not beat the best fixed harness, and its apparent per-task signal is consistent with task memorization; we scope this precisely to the tested predictors and splits (§5.3).

Beyond harnesses, the failure pattern we isolate — LLM builders producing plausible, well-documented, behaviorally inert variants — is a generation-reliability phenomenon we expect to recur wherever a model is asked to modify its own program.

2 RELATED WORK

2.1 HARNESS GENERATION FOR LLMs

A growing line of work generates task-specific scaffolding around a frozen model. TTHe evolves text-to-SQL harnesses with an LLM proposer/judge loop (Nie et al., 2026); Self-Harness (Zhang et al., 2026b) shows a model can improve its own harness; JIT-Agent (Zhang et al., 2026a) trains a model to generate a harness per task at inference time; Task-CoEvolve (Miyai et al., 2026) co-evolves tasks and scaffolds. A concurrent 2026 cluster makes harness evolution *behavior-aware from the verification side*: HarnessLens (Xu et al., 2026) allocates rollout budget using behavior-level checks during evolution; HarnessFix (Chen et al., 2026) repairs harnesses from trace-grounded diagnoses; AHE (Lin et al., 2026) verifies each proposed edit against self-declared behavioral predictions; gated semantic quality-diversity search (Luo et al., 2026) evolves populations over LLM-assigned pathology descriptors; Harness Handbook (Wang et al., 2026) attacks behavior localization and outer-loop search.

Our work is complementary but distinct in unit of analysis and question. Table 1 contrasts the two nearest systems. HarnessLens uses behavior as a *signal inside* an evolution loop to allocate compute; we ask whether the resulting *population* is behaviorally diverse as an outcome property,

	This work	HarnessLens	HarnessBank
Unit of study	population outcome matrix	evolution loop	diversity library
Measures population collapse	yes	no	search collapse only
Gate definition	counterfactual traces	attributable-evidence	semantic screening
Gate calibration	pos/neg controls	—	—
Causal isolation of gate	2×2 factorial	gate ablation (single factor)	no
Confirmatory split	9 untouched DBs	—	—

Table 1: Nearest concurrent systems. Behavior-aware verification inside evolution systems is concurrent independent work; our contribution is the population-level outcome measurement, instrument calibration, and the controlled generation-side contrast.

and what generation-side factors cause or repair its collapse. HarnessBank (Luo et al., 2026) documents search-space collapse and maintains a semantic diversity library with gated screening; our conformance gate does not score semantics at all — it verifies mechanically, via counterfactual trace probes, that a declared mechanism executes — and our design isolates the gate’s causal contribution through a factorial contrast rather than reporting end-to-end system gains.

2.2 HARNESS CONDITIONAL UTILITY AND ROUTING

Several recent papers establish that harness quality is conditional on the instance: Gupta et al. (2026) and the non-monotonicity analysis of Cho (2026) document that no single harness dominates across instances. HELIX (Fan & Huang, 2026) shows a portfolio of sibling harnesses beats the best fixed one and lists a harness router as future work; STS (Zhou et al., 2026) routes over six fixed human-designed scaffolds in a closed set; GRASP (Moll et al., 2026) gates skill edits into a library at development time; Kim (2026) learn counterfactual instance-level routing over {no-search, search, abstain}; Liu et al. (2026) route models within a harness. All of these consume a population whose members *actually disagree*; Fan & Huang (2026) comes closest, evaluating a full candidate outcome matrix and measuring portfolio complementarity beyond the best fixed candidate. Our contribution is narrower and different: we measure whether *automatically generated* populations collapse as an outcome phenomenon, diagnose why at the trace level, and isolate the generation-side causes with a controlled factorial.

2.3 BEHAVIORAL DIVERSITY IN LLM POPULATIONS

Diversity of LLM outputs has mostly been studied at the sample level (self-consistency, diverse decoding) or the agent-policy level. The observation that syntactically distinct programs can exhibit identical behavior echoes coverage gaps documented in program synthesis and mutation-testing practice; our contribution is to instantiate and quantify it for AI-generated harness populations, with a mechanism-level taxonomy.

3 PHASE I: MEASURING AND DIAGNOSING THE COLLAPSE (DISCOVERY)

Phase I is a discovery study: its role is to establish the phenomenon, the taxonomy, and the existence of task-side headroom, and to motivate the confirmatory design. None of its results is confirmatory: 24 of its 151 evaluation items overlapped protocol-development measurements, the human control was designed after inspecting development traces, and its judge was a conservative proxy (Appendix A). We report it with those limits stated.

3.1 HARNESSES, OUTCOMES, AND THE THREE METRICS

A **harness** H wraps a frozen target model M : it receives the natural-language task x , issues one or more calls to M , optionally executes intermediate artifacts (e.g. SQL), and returns a final answer scored programmatically. With H_0 the bare single-call baseline and $Y \in \{0, 1\}$ correctness, the **harness marginal value** is $\Delta(x, H, M) = Y(x, H, M) - Y(x, H_0, M) \in \{-1, 0, +1\}$ (+1 repair, −1 harness-induced harm).

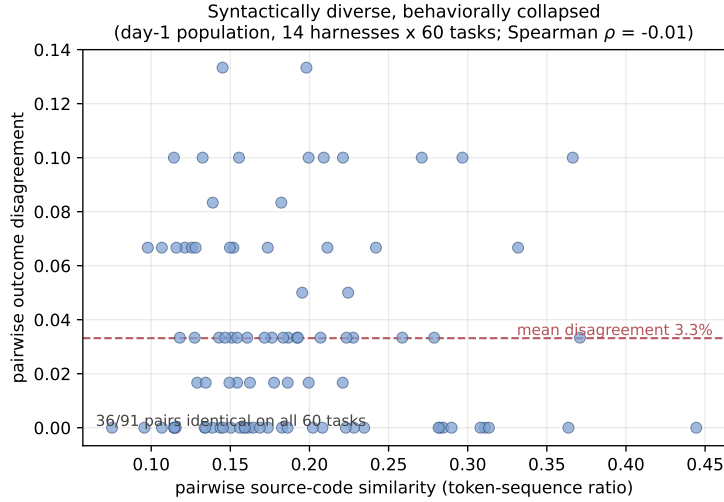


Figure 1: **Syntactic diversity vs. behavioral collapse** (Phase I, discovery; day-1 population: 14 harnesses $\times$ 60 tasks, 91 pairs including the `bare` and `react` baselines). Each point is a harness pair; x = token-sequence code similarity, y = fraction of tasks with differing correctness. Code similarity does not predict outcome disagreement ($\rho = -0.01$); 36/91 pairs are identical on all 60 tasks. Raw pairs: `fig1_phenomenon_data.json` in the code release.

We propose that harness-generation and harness-routing pipelines report three precondition metrics on a held-out set, all computable from the outcome matrix any evaluation already produces: (i) **pairwise outcome disagreement** — over all harness pairs, the fraction of tasks where correctness differs; (ii) **union repair rate** — the fraction of bare errors some population member repairs; (iii) **bare-inclusive oracle headroom**

$$H(P) = \text{Acc}(x \mapsto \max_{H \in P \cup \{H_0\}} Y(x, H)) - \max_{H \in P \cup \{H_0\}} \text{Acc}(H).$$

Positive headroom means population value is *conditional on the item* — the signal a router needs. The metrics are diagnostic in combination: high disagreement with no repair is noise; repair with no headroom means one harness dominates.

3.2 THE COLLAPSE, QUANTIFIED

The domain is BIRD text-to-SQL (SQLite; execution-accuracy judging), target frozen at **GLM-5.3-Flash** (temperature 0) throughout Phase I. On the day-1 population — 14 harnesses (`bare`, `react`, and 12 evolution candidates) $\times$ 60 tasks — average pairwise outcome disagreement is **3.3%** and **36 of 91 pairs are identical on every task**; among the 12 generated candidates alone (66 candidate pairs; mean pairwise code similarity 0.143 under the normalized-character metric of the original code-difference check), disagreement is 3.8% with 21 identical pairs. Every candidate had passed a code-difference check (“all files distinct, non-trivial diffs”). Union repair over bare errors is 7.7% and oracle headroom 3.3 pp; `react` differs from `bare` on zero tasks.

Figure 1 makes the central relation quantitative: across all 91 pairs, pairwise source-code similarity under the figure’s token-sequence metric (mean 0.190; distinct from the candidates-only normalized-character metric of 0.143 above) has Spearman $\rho = -0.010$ with outcome disagreement — code similarity shows *no observed rank association* with behavioral similarity in this population. (Pairs share harnesses and are not independent samples; we report ρ as a descriptive summary, not an inferential test.)

3.3 TRACE TAXONOMY: WHY CODE DIVERSITY FAILS TO BECOME BEHAVIORAL DIVERSITY

Auditing execution traces (the sequence of LLM calls and artifact executions per task) decomposes “different code” into three failure classes:

- **T1 — mechanism described, not executed (6/12 candidates).** The harness issues exactly one LLM call and never executes an artifact — the identical execution path as `bare`. Docstrings claim retrieval, self-checking, or token caps; none exists in the control flow.
- **T2 — feedback without a causal path (`react`, 10/10 audited tasks).** SQL is executed and errors fed back, yet the final SQL is byte-identical to `bare`’s on every audited task.
- **T3 — broken multi-turn path (1/12).** The one candidate implementing genuine multi-turn repair returns prose as final SQL, converting `bare`-correct items into harness-wrong.

The taxonomy is actionable: T1 demands an execution-path acceptance gate, T2 demands verified causal paths for feedback, T3 demands executable-output validation — all generation-protocol requirements.

3.4 HUMAN POSITIVE CONTROL: THE HEADROOM EXISTS

To exclude the alternative that the domain simply lacks instance-conditional signal, we wrote four harnesses by hand with forcibly distinct execution paths (`hpc_repair`: `execute-and-repair`; `hpc_vote3`: 3-vote on execution results; `hpc_schema`: `schema-linking`; `hpc_hint`: `hint-first reordering`). Same frozen target, tasks, and judge as the collapsed population: oracle headroom **6.67 pp** on 60 tasks and **7.28 pp** [3.3, 10.0] on the 151-item set, with `repair/schema/hint` each producing repairs the others miss. One disclosed caveat: the control was designed *after* inspecting day-1 traces, so its headroom is an existence proof, not a blind estimate. The `bare` baseline itself flipped 3 of 60 items between the two day-1 runs (API nondeterminism at temperature 0).

3.5 PHASE-I PROTOCOL PROBES AND THEIR LIMITS

Phase I also probed generation-side fixes — a strategy-forced free-form protocol (assigned mechanism, import/smoke acceptance) and a compiled Harness IR. Under the strategy-forced protocol two of three builders produced populations whose point estimates exceeded the pre-specified thresholds (Qwen3.8-Flash 5.96, DeepSeek-V4-Flash-Vision-Exp 7.95 pp; the old-protocol arm missed at 4.64 pp), and per-strategy acceptance ordered with builder identity (GLM-5.3-Flash 3/8, Qwen3.8-Flash 7/8, DeepSeek-V4-Flash-Vision-Exp 8/8, under the logged tool). These observations motivate Phase II but are not confirmatory, for three disclosed reasons: the design was two nested contrasts that cannot separate protocol from builder main effects; the early runs lack per-attempt logs, so budget equality cannot be audited; and acceptance was an import/smoke check, not the counterfactual conformance gate of §4. Full condition details are in Appendix A.

Generation-null failures (M0). Re-running the original optimize loop with Qwen3.8-Flash and DeepSeek-V4-Flash-Vision-Exp builders produced 40 candidates byte-identical to the `bare` baseline source. Identical code trivially yields identical outcomes, so this run provides no evidence about behavioral collapse; it is reported as a distinct, more primitive failure mode — the pipeline produced no distinct population at all. The collapse claims above rest exclusively on the genuinely distinct-source day-1 population.

4 PHASE II: CONFIRMATORY GENERATION STUDY (DESIGN)

Phase II is the confirmatory study. Its protocol was frozen and committed before any confirmatory outcome was observed (one exception, disclosed as deviation D1 in the protocol changelog: an 18-item pipeline smoke test whose aggregate accuracies were observed and whose items were retained); deviations are logged in the protocol changelog (Appendix B). The frozen estimand, splits, instruments, and analysis plan are summarized here.

4.1 DESIGN

Arms. Two crossed factors: *strategy forcing* ($S=1$: the builder is assigned a named mechanism from a fixed set of eight; $S=0$: the builder invents an improvement with no mechanism named) and the *conformance gate* ($G=1$: the harness must pass the frozen conformance suite; $G=0$: neutral validity only). This yields four factorial arms, which we denote **II-A** (free/ungated), **II-B** (free/gated),

II-C (forced/ungated), **II-D** (forced/gated). A fifth, pre-specified *secondary* arm **II-E** (open mechanism discovery: the builder emits a self-specified behavioral contract over generic trace primitives, gated on its own declaration) tests whether mechanism diversity requires author-supplied vocabulary at all; it is deliberately outside the factorial. These arm names are Phase-II-specific; Phase-I conditions in §3.5 are always referred to with their descriptive names.

Equal raw generation budget. Every slot in every arm receives exactly $R=3$ raw builder attempts (same model, temperature 0.7, 16,384 token cap). Arms differ *only* in which raw candidate is admitted: ungated arms take the first candidate passing neutral validity (parses, imports, returns SQL on a development question — mechanism-blind); gated arms take the first additionally passing the conformance suite. Slot failure is final; populations are never topped up, so generation reliability is part of the treatment. All raw candidates are retained and generation reliability is reported separately from population quality via three rates: R_{artifact} (valid artifact), R_{contract} (valid, non-vacuous contract), R_{fidelity} (asserted properties verified, conditional on a valid contract).

Builders and seeds. Six builders from independent labs (GLM-5.3, Qwen3.8-Max, DeepSeek-V4-Pro, Kimi-K3, MiniMax-M3, ERNIE-5.0-Thinking-Preview), three seeds each: 18 paired (builder $\times$ seed) cells. These were the strongest models available on the project’s single API gateway; no Anthropic- or OpenAI-family builder was accessible, so frontier capability-dependence is untested. Every cell enters the primary contrast regardless of admitted population size, including $K=0$ — excluding low-yield cells would condition on a post-treatment outcome.

Data isolation. A fail-closed contamination registry scans every project artifact for observed BIRD questions. Two databases touched during development are banned wholesale; the confirmatory evaluation uses **1,169 questions from nine databases never opened during development** (the registry’s defensible claim is no project-specific adaptive use, not pretraining-time isolation; the 18-item smoke contact is the disclosed exception), with a stratified 400-item core drawn before any outcome existed. Factorial secondary analyses run on the same core items for all four arms.

Judge. Primary: the official BIRD execution scorer (set equality on raw result rows, 30 s wall-clock timeout). Secondary: the Phase-I legacy judge, measured to be uniformly conservative (on Phase-I matrices it never credits an answer the official scorer rejects).

Instruments. The conformance suite drives each harness against scripted scenarios and asserts on the recorded execution trace, never on task performance, so gating cannot leak outcome supervision. Every mechanism is probed with *paired counterfactual variants* whose correct answer differs — including a variant where the first query *succeeds*, separating conditional repair from an unconditional second call — and answers are compared by exact normalized equality. The suite was calibrated before any Phase-II generation against three hand-written positive controls (all must pass) and two known-negative impostor harnesses (both must be rejected), plus the invariant that the bare baseline must violate every conditional property; the calibration is an executable gate re-run before each generation batch. Arm II-E additionally enforces non-vacuity (the contract must be violated by the bare baseline, checked by running it) and verifies only the properties actually asserted.

4.2 FROZEN ANALYSIS PLAN

The **primary estimand** is the paired difference in bare-inclusive oracle headroom, $H(\text{II-D}) - H(\text{II-A})$ over (builder $\times$ seed) cells, on the full 1,169-question set under the official judge — a continuous effect with a two-sided 95% CI and paired permutation test, never a pass/fail threshold. The Phase-I thresholds survive only as a descriptive checklist. **Confirmatory secondaries** (Holm-corrected, all four arms on the same core-400 items): gate main effect $[(B-A)+(D-C)]/2$, strategy main effect $[(C-A)+(D-B)]/2$, and their interaction $(D-C)-(B-A)$ — whether strategy differentiation requires execution fidelity to be realized. **Inference:** hierarchical paired bootstrap (10,000 replicates; databases resampled, then tasks within database, cells paired). $H(P)$ is *not* monotone in K — adding a harness that answers everything correctly drives headroom to zero — so matched- K comparison, not monotonicity, licenses cross-population claims. K -matched sensitivity, the full estimand definitions, and the robustness audits (dual judge, cache-off replication, repeat stability) are specified in Appendix B together with the protocol deviation log.

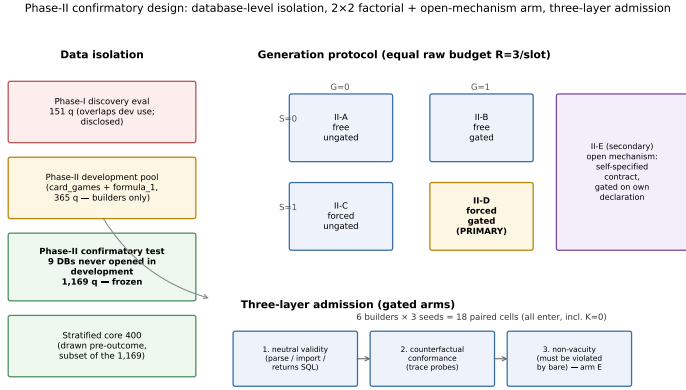


Figure 2: **Phase-II confirmatory design** (frozen protocol). Left: database-level contamination isolation (the two development databases are banned wholesale; evaluation uses nine databases never opened during development). Middle: the 2x2 factorial (strategy forcing x conformance gate) with equal raw budgets, plus the open-mechanism arm II-E. Right: the three-layer admission pipeline (neutral validity → counterfactual conformance → non-vacuity vs. bare).

	Arm II-A (free/ungated)	Arm II-D (forced/gated)
Oracle headroom $H(P)$, official judge	8.6 pp (cell mean)	8.4 pp (cell mean)
Union repair rate	31.3%	30.0%
Union harm rate	19.8%	24.6%
Pairwise disagreement	10.4%	12.6%
Unique outcome-vector fraction K_{eff}/K	1.00	1.00
Paired contrast $H(\text{II-D}) - H(\text{II-A})$, 18 cells:		
Point estimate	-0.16 pp	
Two-sided 95% CI (hierarchical bootstrap)	[-0.90, +0.82] pp	
Paired permutation p (one-sided)	0.65	

Table 2: **Primary result.** Bare-inclusive oracle headroom per arm and the paired D–A contrast, pooled over 6 builders x 3 seeds on the 1,169-question confirmatory set (nine databases never opened during development), official BIRD judge. The CI spans zero: **the combined intervention shows no clear improvement**. Per-cell values are in Figure 3 (left).

5 PHASE II: RESULTS

5.1 PRIMARY CONTRAST: FORCED+GATED VS. FREE+UNGATED

The confirmatory answer to our primary question is negative: the strategy-forced, conformance-gated protocol does not increase population headroom relative to free, ungated generation. The point estimate is small (−0.16 pp), the hierarchical bootstrap CI comfortably spans zero, and the permutation test offers no evidence against the null. Per the pre-specified conditional conclusions, we report this as *no clear improvement* — an unresolved estimate, not evidence of no effect.

Two secondary observations qualify the null. First, both arms carry *substantial* headroom (A: 3.1–11.7 pp across cells; D: 4.5–11.9 pp): under the clean protocol, free generation alone produces headroom-rich populations. Second, the per-cell deltas are heterogeneous (−2.5 to +2.8 pp), small-K gated cells trending positive and large-free cells negative — though the noise-floor analysis below cautions against reading this spread structurally. Arm D repairs marginally fewer bare errors (30.0% vs. 31.3%) while inducing more harm (24.6% vs. 19.8%).

5.2 FACTORIAL DECOMPOSITION (SAME CORE-400 ITEMS)

The factorial decomposition must be read with two corrections the frozen plan anticipated and the data now supply.

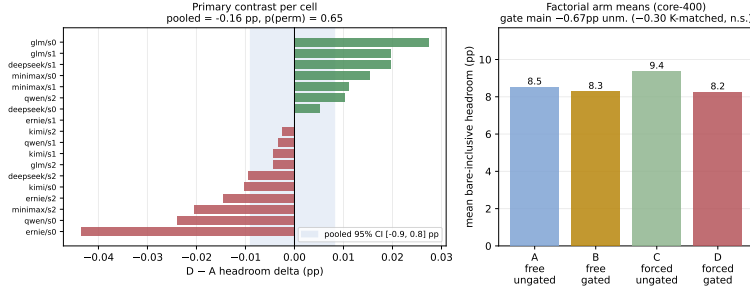


Figure 3: **Confirmatory results.** Left: per-cell D-A headroom deltas (sorted; green = D better, red = A better) with the pooled 95% CI band. Right: factorial arm means on the identical core-400 items; the conformance gate’s negative main effect is starred (CI excludes zero). Both panels derive from the frozen analysis scripts.

Contrast	Effect on headroom, 95% CI
Gate main effect $[(B-A) + (D-C)]/2$	-0.67 pp [-1.17, -0.13]
Strategy main effect $[(C-A) + (D-B)]/2$	+0.41 pp [-0.22, +1.11]
Strategy $\times$ gate interaction $(D-C) - (B-A)$	-0.87 pp [-2.22, +0.31]

Table 3: **Confirmatory secondaries**, 18 quad cells, all four arms on the identical 400-item core. The gate main effect is the only contrast whose CI excludes zero — on the negative side.

Size confounding. The gate admits $\sim 30\%$ fewer candidates (69 vs. C’s 101 slots), and oracle headroom grows with population size, so part of the unmatched -0.67 pp is mechanical. At **matched population size** (subsampling each ungated population to its gated arm’s K , 200 draws), the gate main effect shrinks to -0.30 pp, 95% CI $[-0.73, +0.11]$ — *not distinguishable from zero*. More than half the unmatched effect was population size.

Multiplicity. The three factorial contrasts form a pre-specified Holm family. Raw permutation p for the gate main effect is 0.028, but the **Holm-adjusted** p is 0.084 — not confirmatory even before K-matching. The unmatched CI excluding zero was never pre-specified significance.

Noise floor. Three independent cache-off repeats of bare (R3, temperature 0) flip 14.5% of task outcomes at least once (pairwise ~ 9.7 –10.0%; bare accuracy varies 63.8–65.8% across repeats). The cache-off arm replication (R2; arms A and D, 70 harnesses, 28,000 core cells) flips 8.5% of verdicts against the cached run while leaving arm-level disagreement structure and every subset contrast unchanged (Appendix B): same-harness rerun disagreement is of the same order as cross-harness disagreement, so a substantial share of measured pairwise diversity is run noise — and the sub-pp effects under interpretation sit an order of magnitude below this floor.

Headroom decomposition. The gate does not make populations worse: oracle accuracy is essentially unchanged (A .761 vs. D .757; Table 4), and the unmatched headroom delta is carried by best-fixed accuracy shifting, not by a loss of union correctness. “Less headroom” here means *less routing elbow-room*, not a worse population.

The honest summary: neither component moved population headroom beyond noise in either direction; we report the unmatched, K-matched, adjusted, and decomposition views side by side and decline to interpret the residual.

5.3 DIVERSITY DOES NOT IMPLY LEARNABLE ROUTING

Open-set value prediction does not beat the best fixed harness. A leave-one-harness-out predictor over the Phase-I admitted populations fails to beat the best fixed harness (DeepSeek-V4-Flash-Vision-Exp arm -3.4 pp, Qwen3.8-Flash $+0.9$ pp, CI $[-5.2, +6.9]$; cross-protocol pool -0.9 pp). Apparent per-fold repair AUROC of 0.82–0.98 collapses under database-stratified task-held-out controls (mean 0.512) — the signal is task memorization, not harness-value estimation. One narrow positive survives: as an *abstaining gate* on the 24-harness cross-protocol pool, the same predictor

Arm	K	bare	mean H_i	best-fixed	oracle	headroom
A free/ungated	6.8	.653	.658	.676	.761	.086
B free/gated	7.1	.632	.642	.661	.744	.083
C forced/ung.	6.6	.632	.624	.656	.750	.094
D forced/gated	4.8	.653	.646	.673	.757	.084

Table 4: **Headroom decomposition** (18 quad cells, core-400, official judge). Oracle accuracy is essentially unchanged across arms; the unmatched headroom differences are carried by best-fixed accuracy, not by population utility. Yield: $K/144$ slots A/B/C/D = 104/110/101/69; per-builder R_{artifact} 0.13 (GLM) to 0.80 (ERNIE).

gained +6.9pp over bare with zero *observed* induced harm — on its own tuning tasks, so not a deployment claim. Selective abstention may be easier than full routing; per-item value estimation from code and task text alone remains open.

5.4 EXPLORATORY: WHAT TO MEASURE, AND WHAT TO SELECT FOR

Two post-hoc analyses on the archived confirmatory cells (details in Appendix C) answer the two questions a measurement paper invites. **(i) Output-space similarity carries the signal that code-space similarity lacks:** mean final-SQL dissimilarity on verdict-*agreeing* cells correlates with outcome disagreement at Spearman $\rho=0.59$ (within-arm 0.57), where code similarity showed none ($\rho=-0.01$) — but richer channels are noisier channels (same-harness reruns change call counts on 1.8% of cells, verdicts on 8.5%, final-SQL text on 70%), and forced-mechanism arms cost 39–42% multi-call trajectories (free arms 25–30%) while scoring *lower* (.62 vs. .64). **(ii) Diversity-aware population selection transfers:** selecting $k=8$ per cell to maximize dev-half oracle coverage yields +0.82 pp held-out headroom over top-by-accuracy selection (95% CI [0.43, 1.31]) and +0.40 pp over random ([0.18, 0.66]); accuracy selection is itself worse than random — it buys the best *member*, diversity buys *complementarity*. Population-level statistics transfer across the database split; per-item value estimates do not (§5.3).

6 DISCUSSION AND CONCLUSION

Protocol sensitivity is the finding. The Phase-I collapse did not reproduce under the clean equal-budget protocol: free ungated generation already produced 8.6 pp headroom and 31% union repair. Collapse is a failure mode of *specific* pipelines, invisible to every code-level check they passed. Mechanism-verification added no routing value, its apparent cost largely a population-size artifact: outcome-level auditing, not stronger code-level gating, detects collapse. Sub-pp effects sit below the repeat-run noise floor (14.5% bare any-flip), which belongs beside every CI. The decompositions show the gate changed *which* candidates survive, not *whether* the population can answer: reliability and complementarity need separate reporting.

Limitations. One domain family (text-to-SQL); Phase-I from a discovery design with disclosed task overlap, a proxy judge, and a post-trace human control; inference conditional on the six sampled builders; early Phase-I generations lack per-attempt logs; the analyses of §5.4 are exploratory (post-hoc on confirmatory outcomes). A one-seed MATH-500 free-arm replication (Appendix G) reproduces the clean-protocol diversity result in a second domain — including the sharp separation between population-level complementarity (76% union repair) and member-level utility (no generated harness beats bare) — but is exploratory and does not test the collapse-inducing protocol.

AI-generated harness populations can be exactly what their source files suggest and nothing like what routing needs — and the reverse can also hold. Phase I measured the syntactic-behavioral gap, traced its mechanisms, and showed the missing headroom exists on the task. A pre-specified confirmatory factorial study (six builders, three seeds, 2×2 forcing $\times$ gate, 1,169 unseen questions) then revised the story twice. **Collapse proved protocol-sensitive:** a failure mode of specific pipelines, invisible to every code-level check they passed. **The intervention was null within measurement precision,** its sub-pp effects sitting below the noise floor. The practical output stands: measure disagreement, repair, harm, and headroom before routing, select populations for complementarity rather than accuracy, and audit the generation protocol itself.

AI USE STATEMENT

This work is, by construction, an AI-assisted research project studying AI-generated artifacts, and we disclose the extent of that assistance precisely.

Required disclosure. We used generative AI tools for: (i) writing and modifying the experimental code (generation pipeline, evaluation collectors, statistical analysis scripts), which was executed and verified through automated tests, calibration against hand-written positive controls and known-negative mutants, and cross-model adversarial review; (ii) executing experiments, including all model API calls reported; (iii) analyzing experimental data and producing tables and figures; and (iv) drafting and editing manuscript text. Every quantitative claim in this paper is generated by committed, version-controlled analysis scripts from archived raw outputs, and every deviation from the pre-frozen protocol is disclosed in the protocol changelog.

Research ideas and direction. The core research questions, the causal design decisions (estimand, contrasts, exclusion rules), the interpretation standards, and all go/no-go decisions were made by the human authors, with AI tools acting as executor and as one adversarial reviewer (a different model family from the executor, per our cross-model review discipline).

Responsibility. We take responsibility for the final content of this work, including text, claims, and artifacts produced with the aid of generative AI. The factual statements in this section are provided for author verification; final responsibility for their accuracy rests with the authors.

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A PHASE-I PROVENANCE AND CONDITIONS (DETAIL)

Discovery status. Phase I (Sections 3 and 5.3) is a discovery study. 24 of its 151 evaluation items overlapped day-1 protocol-development measurements; the human control was designed after inspecting day-1 traces (its headroom is an existence proof, not a blind estimate); the repository history was initialized retrospectively, and the decision-log changelog is the authoritative ordering record. Phase-I admission thresholds (union repair $\geq 15\%$, headroom ≥ 5 pp, ≥ 3 pairwise-distinct non-empty fix sets) were frozen in the internal decision log before the 151-item evaluation; the v6 $\rightarrow$ v7 threshold change (20% $\rightarrow$ 15%) is disclosed, and both readings admit the same populations. This is an internal pre-evaluation freeze, not external preregistration.

Data hygiene. The bare baseline flipped 3 of 60 items between the two day-1 runs (API nondeterminism at temperature 0), so 60-item columns are conditioned on slightly different bare-error sets. Phase-I judging used a set-of-rows-equality proxy that we later measured to be uniformly conservative relative to the official BIRD scorer (on the two Phase-I matrices with logged final SQL, it never credits an answer the official scorer rejects; agreement 0.97–0.99).

Phase-I populations. Seed baselines: *bare* (single call) and *react* (execution feedback loop, shipped with TTHE). *Old proposer protocol*: six candidates selected from TTHE’s optimize-loop output (GLM-5.3-Flash builder), selected for behavioral diversity using only the 36 questions disjoint from the evaluation set, by a rule frozen before evaluation. *Strategy-forced free-form*: GLM-5.3-Flash, Qwen3.8-Flash, and DeepSeek-V4-Flash-Vision-Exp builders, eight assigned mechanisms each; the GLM population is the logged re-run ($K=3$; the original 09-01 run recorded zero accepted outputs but its tool was not preserved and the observation is unaudited). *Harness IR*: six harnesses compiled deterministically from declarative mechanism specifications (context $\in \{\text{plain, schema.link, hint.guard, link+hint}\}$, candidate generation with temperature, per-candidate execution, repair rounds, selection rule), so a declared mechanism is present in the control flow by construction. *Human positive control*: the four hand-written harnesses of §3.4.

Generation-provenance gaps. The 09-01 free-form populations were produced in-session; their generation tool was not preserved, and per-attempt logs do not exist for them — so budget equality cannot be audited for those runs, and we do not claim it. Their acceptance was an import/smoke check at evaluation time, *not* the Phase-II counterfactual conformance gate. The tool of record from 2026-09-03 (*builder_generate3.py*) logs every attempt (3 retries, 16,384-token budget,

per-attempt stage and finish reason); the completed per-strategy grids and the builder-acceptance ordering (3/8, 7/8, 8/8) come from this logged tool and are post-hoc with respect to the frozen admission decisions.

B PHASE-II PROTOCOL SUMMARY AND DEVIATION LOG

The authoritative protocol (estimands, splits, instruments, exclusion rules, and the complete deviation changelog D1–D13) is `PHASE2_PROTOCOL.md` in the code release, frozen at the commit recorded in `PROTOCOL_FREEZE.txt`. Summary: 2×2 factorial plus open-mechanism arm; six builders $\times$ three seeds; equal raw budget $R=3$ per slot with no top-up; contamination registry (fail-closed, two development databases banned wholesale); 1,169-question confirmatory set from nine databases never opened during development plus a pre-drawn 400-item core; official BIRD judge with enforced 30 s timeout; response cache keyed on model identity with cross-process locking (deviation D13 documents and fixes a cache concurrency bug found during collection); primary contrast D–A on paired (builder $\times$ seed) cells with hierarchical bootstrap and permutation inference; Holm-corrected factorial secondaries on the common core; K-matched sensitivity only. Deviations D1–D13 include the disclosed 18-item pipeline smoke test, instrument-calibration fixes made before any confirmatory outcome was observed, the credential-handling incident and its remediation, and the artifact-divergence quarantine — each with its rationale and date in the changelog.

R2: cache-off arm replication (detail). Arms A and D were re-run cache-off on the 400-item core: 70 harnesses (the first two admitted per builder $\times$ seed cell), 28,000 cells. Results: 8.5% of cell verdicts flip against the cached confirmatory run; two independent cache-off passes over arm D (a duplicate collection, disclosed as a scheduling artifact and retained as an independent replicate) flip 8.7% against each other. Arm-level structure is unchanged — pairwise disagreement A 10.4% $\rightarrow$ 10.3%, D 14.3% $\rightarrow$ 14.4%; subset headroom A 4.1 $\rightarrow$ 3.6 pp, D 4.0 $\rightarrow$ 4.0 pp; mean subset D–A -0.1 pp cached vs. $+0.4$ pp cache-off. Same-harness rerun disagreement is of the same order as cross-harness disagreement, so a substantial share of measured pairwise diversity is run noise; the shared cache neither manufactured nor masked the diversity signal.

C EXPLORATORY FINGERPRINT AND SELECTION ANALYSES

Both analyses are post-hoc on the archived confirmatory cells (18 builder $\times$ seed cells, 384 harnesses, 4,386 within-cell pairs, core-400 outcomes; scripts `w3_fingerprint.py` and `w1_selection.py`, outputs `w3_fingerprint.json` and `w1_selection.json`). They are exploratory: the confirmatory set had already been analyzed, and no result here was pre-specified.

Rich fingerprints. SQL text is normalized by case-folding outside string literals (literal case is verdict-relevant: `'Eur'` vs. `'EUR'` flips the official judge), quote and semicolon stripping, and whitespace joining. Consistency check: among 365,095 cells where a pair emits identical normalized SQL, verdict disagreement is exactly zero — the artifact channel is verdict-consistent by construction under a deterministic judge. Pair-level artifact dissimilarity is the mean (1–token-Jaccard) over *verdict-agreeing* cells only, so it is not mechanically inflated by disagreement cells; its Spearman correlation with overall outcome disagreement is 0.59 (all within-cell pairs) and 0.57 (the 1,010 within-arm pairs). Rerun stability is measured on the R2 cache-off replication (70 harnesses, 28,000 cells): call-count changes 1.8%, verdicts 8.5%, normalized SQL 70.0% (mean token Jaccard 0.62 on changed cells) — the richer the channel, the less reproducible a single run of it is, which is why the verdict channel anchors the paper’s metric and the richer channels enter only as correlates. Per-arm cost: mean multi-call fraction A .298, B .245, C .422, D .392; mean accuracy A .641, B .644, C .620, D .620.

Diversity-aware selection. Per cell, the pool is all admit-confirmed harnesses of that cell (arms mixed). Tasks are split by database halves (100 random splits); selection uses dev only, evaluation held-out only. Methods: *top-acc* (top- k dev accuracy), *random* (mean over 200 draws), *div-only* (greedy maximization of dev oracle coverage), *acc+ λ div* (greedy dev mean-accuracy + λ ·oracle). Inference is a cell-level bootstrap (18 cells) over split-averaged paired deltas. At $k=8$: div-only vs. top-acc $+0.82$ pp [0.43, 1.31], vs. random $+0.40$ [0.18, 0.66], random vs. top-acc $+0.41$

[0.19, 0.68]; at $k=4$: +0.65 [0.23, 1.17], +0.24 [−0.03, 0.53] (not distinguishable from random), +0.41 [0.11, 0.74]. The λ -blended variants interpolate monotonically between top-acc and div-only; the pure diversity objective dominates at both k . A dev-measured population statistic (oracle coverage) transfers across databases; a dev-measured per-item signal does not (§5.3) — which is the operational content of the measurement mandate.

D PHASE-I ADDITIONAL RESULTS

Per-harness accuracies, per-fold LOHO tables, bootstrap intervals, fix-set audits, K-subsampling curves, the task-held-out control, and the second-target replication (Qwen3.8-Flash as frozen target: collapse replicates; the strategy-forced population retains headroom; the prompt-side human control does not transfer wholesale) are included in the anonymous code release with their generating scripts.

E MODELS AND VERSIONS

Table 5 lists every model used in this paper with its exact version string as sent to the API. All models were accessed through a single commercial OpenAI-compatible API aggregator (the same provider for every role); the data-collection window is 2026-08-30 through 2026-09-08. The frozen target is always called at temperature 0. Phase-II harness generation uses temperature 0.7 with a 16,384-token budget; Phase-I generation settings are recorded per tool in the code release. All correctness verdicts reported in both phases come from the deterministic official BIRD execution-accuracy scorer (30 s timeout); no LLM judge produces any reported accuracy.

Role	Model (exact version string)	Used in
Frozen target solver	GLM-5.3-Flash (temperature 0)	Phases I–II
Second frozen target (replication)	Qwen3.8-Flash	Phase I
Phase-I builders	GLM-5.3-Flash, Qwen3.8-Flash, DeepSeek-V4-Flash-Vision-Exp	§3.5
Phase-II builders	GLM-5.3, Qwen3.8-Max, DeepSeek-V4-Pro, Kimi-K3, MiniMax-M3, ERNIE-5.0-Thinking-Preview	Phase II
Judge	official BIRD scorer (deterministic execution accuracy)	all results

Table 5: Complete model inventory. The target is held frozen within each phase; Phase-II builders come from six independent labs.

F WORKED EXAMPLES FROM THE PHASE-I TRACE AUDIT

To make the failure taxonomy of §3.3 concrete, this section walks one BIRD task through the audited harnesses. Artifacts are verbatim from the day-1 trace audit (5 harnesses $\times$ 10 tasks); its logger caps stored SQL excerpts at 120 characters, so truncations are marked *[trunc.]*. Full harness sources and traces are in the code release.

The task. BIRD dev-set question 352 (*card_games* database; question text verbatim, the typo is the dataset’s):

Calculate the percentage of the cards availabe in Chinese Simplified.

Its BIRD hint:

```
'Chinese Simplified' is the language; percentage =
Divide(Sum(id where language = 'Chinese Simplified'),
Count(id)) *100
```

Gold SQL (executing the two on the released database yields 8.7734 and 8.7728 respectively — the official scorer separates them):

```
SELECT CAST(SUM(CASE WHEN T2.language = 'Chinese Simplified'
THEN 1 ELSE 0 END) AS REAL) * 100 / COUNT(T1.id) FROM cards
AS T1 INNER JOIN foreign_data AS T2 ON T1.uuid = T2.uuid
```

The bare harness answers:

```
SELECT SUM(CASE WHEN language = 'Chinese Simplified'
THEN 1 ELSE 0 END) * 100.0 / COUNT(id) AS percentage FROM
foreign_data;
```

marked wrong by the official scorer: it denominates by the number of `foreign_data` rows (one per translated language per card), while the gold divides by the number of cards — a join the hint’s phrasing does not spell out.

react (T2: feedback without a causal path). `react`’s control flow does execute SQL and retries on database errors. But its first call is byte-identical to `bare`’s prompt, and here the generated SQL executes without error, so the error-triggered retry never fires and the final SQL equals `bare`’s. Across all 10 audited tasks the pattern held: SQL executions present, final SQL byte-identical to `bare`’s on 10/10 — the feedback loop is real code with no behavioral consequence.

cand.bird.g1.b0r1.g0 (code-level diversity, behavioral identity). This candidate is a genuinely different program: it wraps the single coder call in a daemon-thread watchdog and fires one hedged duplicate re-ask if the call stalls past 40 s. The mechanism exists in the control flow — but its trigger (a stalled API call) never occurred in the audit: zero SQL executions, and the final SQL equaled `bare`’s on all 10 audited tasks. Code difference without behavioral difference survives even when the added mechanism is real: it simply never fires.

cand.bird.g1.b1r1.g1 (T3: a broken multi-turn path). The one audited candidate implementing genuine execute-inspect-repair returns the model’s *prose* as final SQL. On a task `bare` answered correctly (“State the alternative languages available for card named Annul numbered 29.”), its logged final SQL begins, verbatim:

```
with "only a single read-only SELECT is allowed". The
previous "query" was actually just prose text (the model
wrote rea[trunc.]
```

The repair re-prompt invites the failure: asked to fix a query that “was actually just prose”, the frozen target responds with more prose reasoning, no SQL block is found, and the reasoning text is submitted as the query. The item flips from `bare-correct` to `harness-wrong`.

G CROSS-DOMAIN REPLICATION: MATH-500 MINI-AUDIT

To test whether the Phase-II clean-protocol finding (free ungated generation produces behaviorally diverse, complementary populations) is text-to-SQL-specific, we replicated the free-arm protocol on competition math.

Protocol. Six builders (the same models as Phase II), one seed, eight slots each, $R=3$ raw attempts per slot, neutral validity gate (mechanism-blind: parse, import, interface, non-empty answer on two dev questions). Frozen target GLM-5.3-Flash (same as the SQL study), temperature 0. MATH-500 (Lightman et al., 2023): 500 tasks frozen before generation (100 dev / 400 eval, seed 20260908). Judge: numeric equality where both sides parse as numbers (including a/b , $\frac{a}{b}$, decimals), else normalized LaTeX string equality (units, base subscripts, ° stripped; X content preserved; $\text{pmatrix} \rightarrow \text{tuple}$). Judge self-consistency: 400/400 gold answers judge correct against themselves. `bare` accuracy 0.938 under the matched answer-format instruction. Yield: 35/48 slots admitted (R_{artifact} 0.10–1.00; the two reasoning-heavy builders, GLM-5.3 and Qwen3.8-Max, exhausted the 32k-token generation budget on reasoning before emitting code in most slots — reported as-is). This is a replication probe, not a factorial: no strategy forcing, no conformance gate.

Results (35 harnesses, 400 eval tasks). Mean pairwise outcome disagreement 10.6% (max 33.0%), K_{eff} 33/35 (0.94), outcome-identical pairs 0.7% — no trace of Phase-I-style collapse. Union repair 76.0% of bare errors, union harm 0.8%, oracle headroom over bare 4.75 pp. Yet the best fixed member is *bare itself* (0.9375; generated harnesses 0.68–0.87): the population is strongly complementary at the oracle level while every individual member is worse than the baseline it wraps — diversity is real, utility is at the population level only, and the outcome matrix is what shows both.

Two failure-side observations. First, the clean-protocol diversity result replicates in a second domain with a different output space. Second, a quarantined pilot round (disclosed in `PROTOCOL_CHANGELOG.md`, entry G1) supplies a controlled contrast: when the same six builders were (mistakenly) given a GSM8K-style task description whose numeric answer format did not match MATH-500, the generated population remained behaviorally diverse (16.2% disagreement, K_{eff} 19/19) while its union repair collapsed to 18% — and code-level inspection of the catastrophic members reproduced the Phase-I trace taxonomy verbatim (a generate-then-verify harness whose verifier rejected nearly every answer and whose fallback returned the stale loop variable or “0”). Diversity without utility, induced by a prompt-side protocol error invisible to code-level checks: the measurement mandate, not any code-level audit, is what detects it.

Scope. One seed, one target, no factorial, exploratory; the Phase-I collapse-inducing SQL protocol was not re-run in the math domain, so this replication speaks to the clean-protocol result only, not to the boundary of collapse.